Important Notes

1. All geometry was adjusted to standard slicer settings and basic PLA, you can add walls and infill for stronger parts, using PETG will also enforce parts.

2. In most cases flat bolts can be swapped to countersunk bolts.

3. We used TPU by bambulab for tires, they should print in 7H, standard settings for TPU in bambulab slicer are recommended.

4. Main grey color used in videos is light grey 10104 PLA by bambu.

5. We highly recommend using 2F modules after base and before roof, when you are using modifications.

6. Holes for casters are M8, make sure to check it before purchasing them.

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VAULT ·····

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VAULT



3D Printed Modular Storage System

Instructions Manual

·· 2

····· S/M/L/XL ····

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Hardware List

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DRAWERS

1.	M4 14MM bolts	x4	*per floor*	
2.	M6 50MM bolts	x2	*per roof handle*	

SUSTAINERS

1.	M4 10MM bolts	x8	*per module*	
2.	M4 14MM bolts	x16	*per module*	
3.	M4 20MM bolts	x7	*per module*	

MODIFICATIONS

1.	M6 30MM bolts	x4	*per pull-out handle*	
2.	M6 35MM bolts	x2	*per pull-out handle*	
3.	M6x8x8 heat insert	x2	*per pull-out handle*	
4.	M4 60MM bolts	x8	*per static handle*	
5.	M4 40MM bolts	x4	*per pegboard board*	
6.	M4 60-100MM bolts	x4	*per backbone board*	
7.	M16 504MM rod	x1	*per XL base*	
8.	M16 414MM rod	x1	*per L base*	
9.	M16 338MM rod	x1	*per M base*	
10.	M16 248MM rod	x1	*per S base*	
11.	M16 N-MM rod	x2	*per pull-out handle*	
12.	M4 25MM bolts	x6	*per wheel*	
13.	M6 90MM bolts	x2	*per foot*	
14.	6205ZZ bearing	x1	*per wheel*	
15.	SCSAW stopper	x1	*per wheel*	

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Five Ways Of Using - VAULT

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3D·····

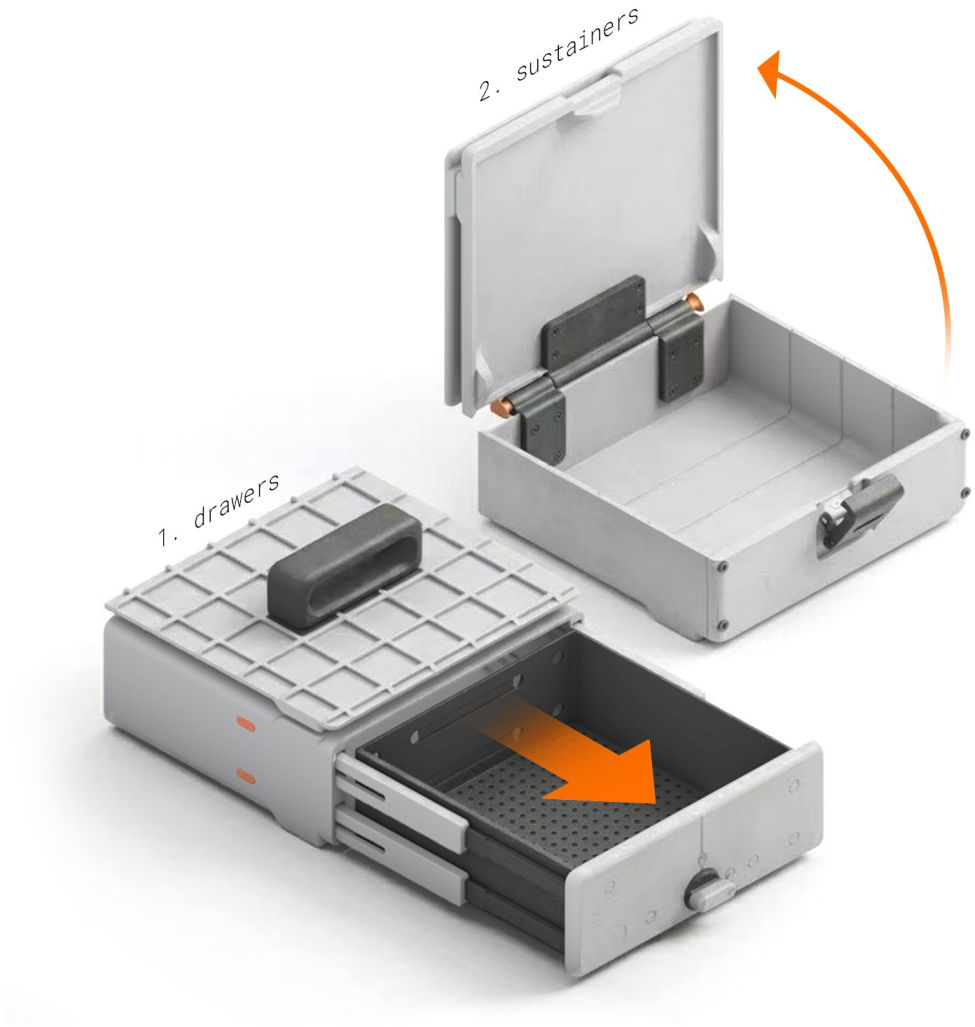
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Two Types of - VAULT

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*archive is split into two parts as well

·· 5

42mm Gridfinity ·····

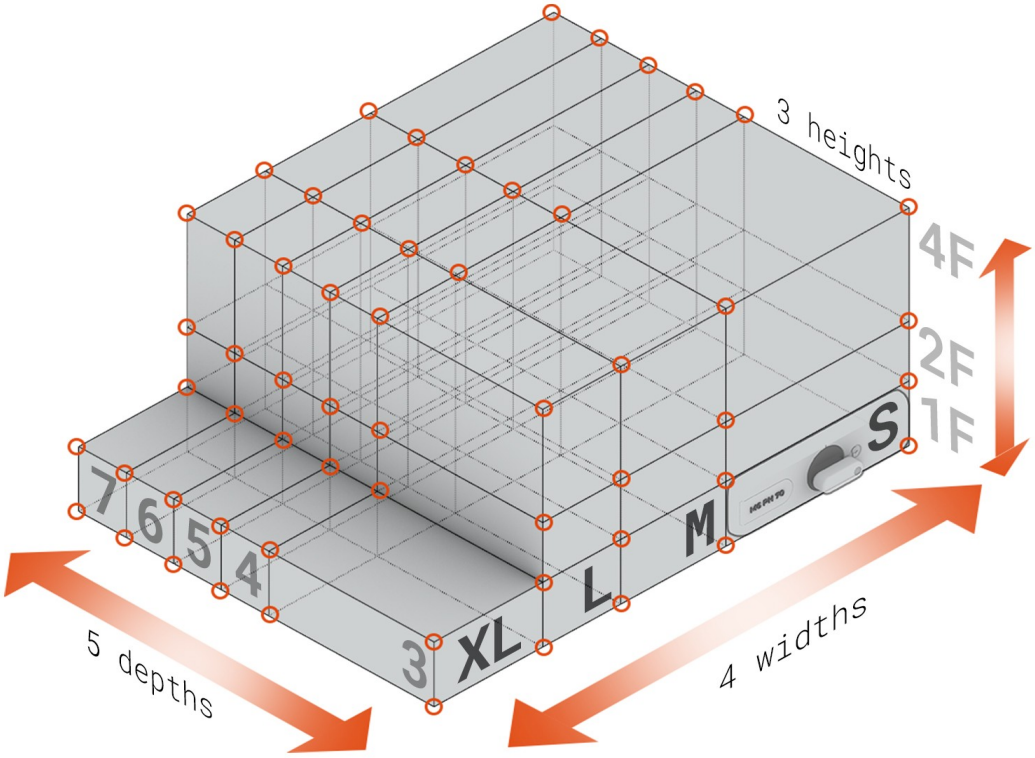
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S, M, L, XL - 50 Size Variants

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*depths are determined by gridfinity units (42mm x3, 42mm x4..)
*widths are determined by gridfinity units and printer sizes
*1F, 2F and 4F are the heights, no 3F to avoid gaps, everything divides by 2

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····· 1F/2F/4F ····

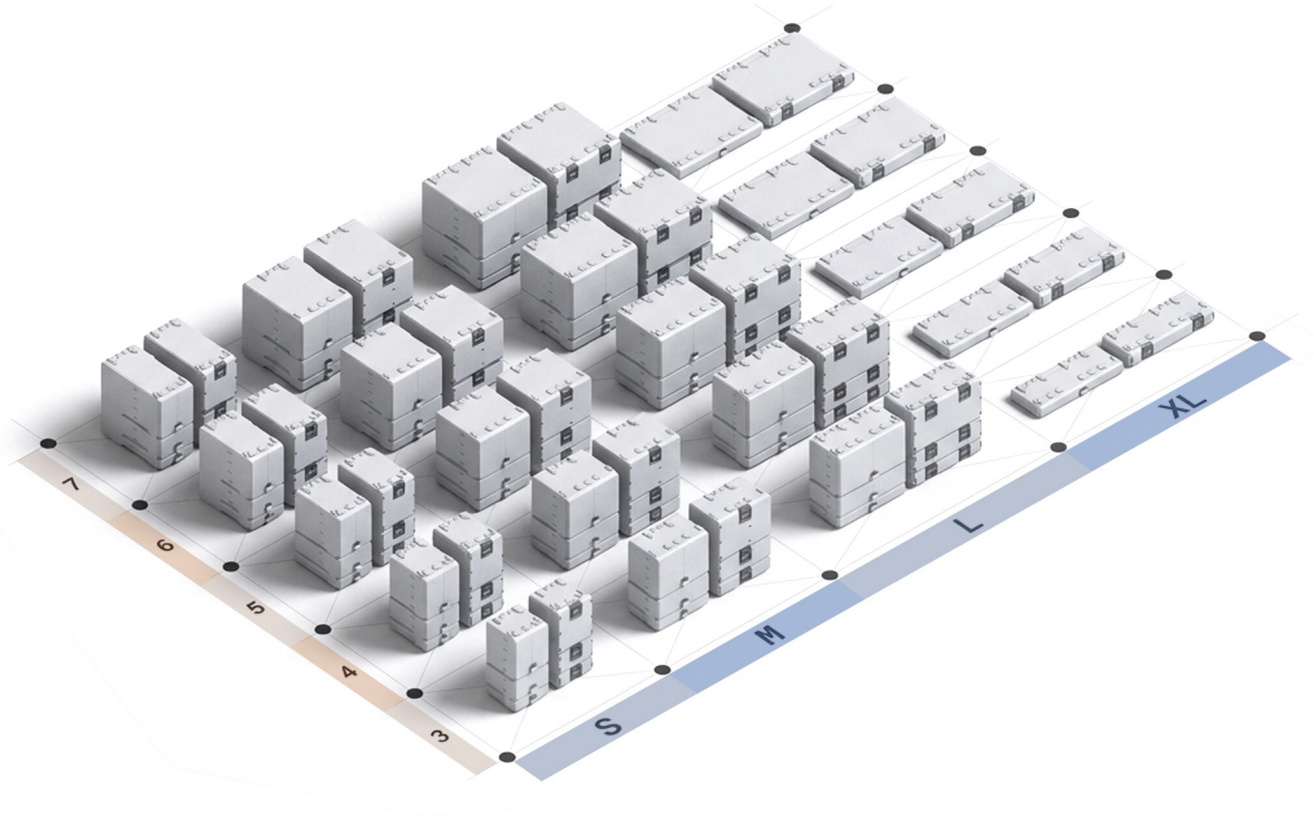
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160 Module Variants

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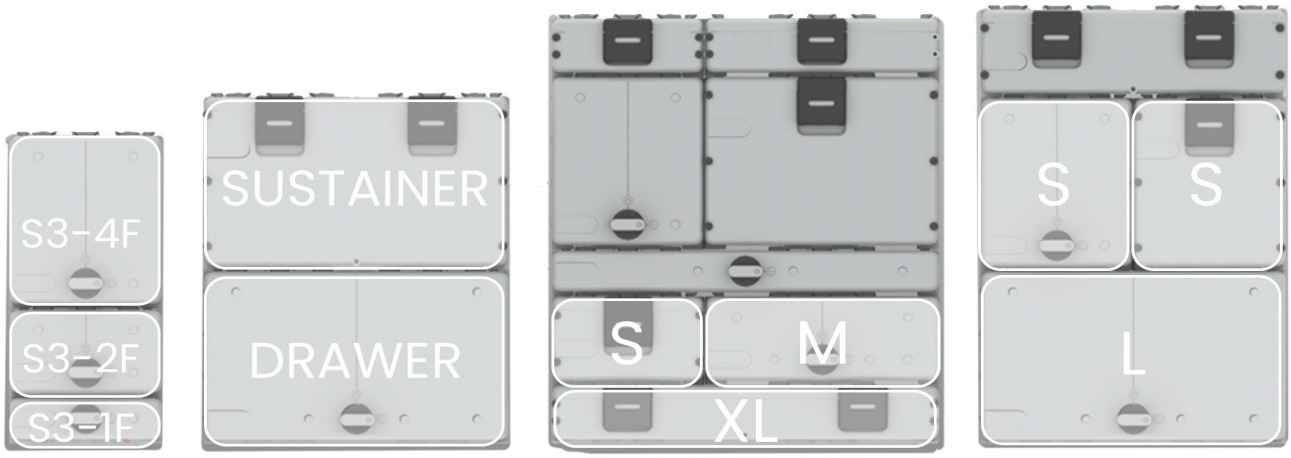
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Stacking Options

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1. Modules with similar depth and width stack on each other.
2. SUstainers and Drawers can stack as well.
3. XL in width, equals to S+M, so they stack as well.
4. L in width, equals to S+S. they stack too.

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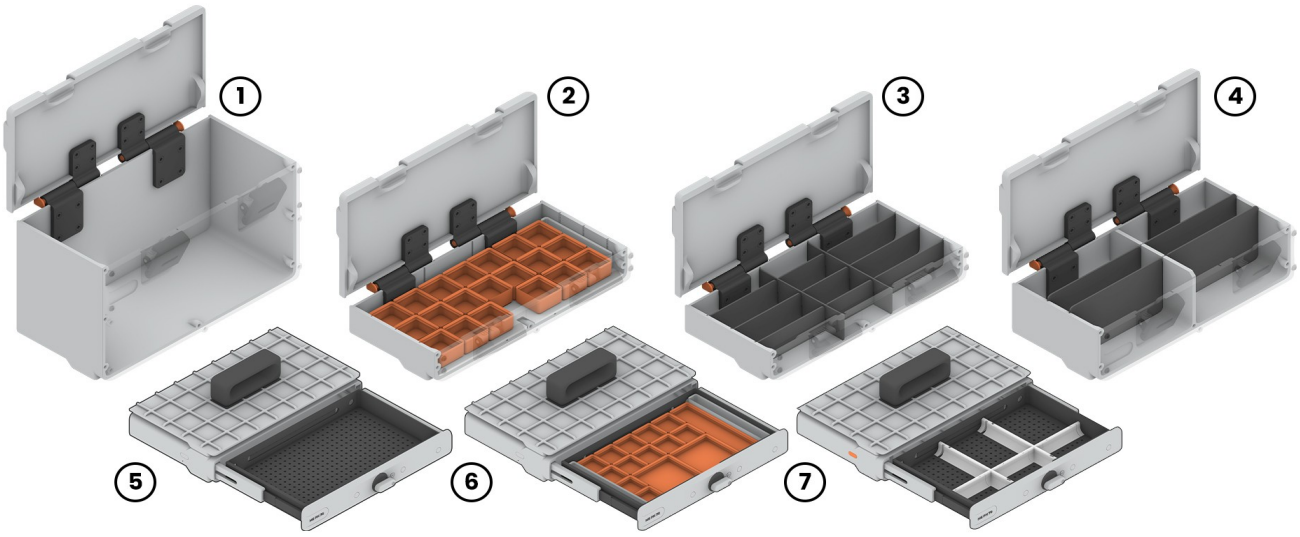
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Interiors

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1. flat sustainer 2. gridfinity base 3. dividers sustainer 4. sealed sustainer
5. empty drawer 6. gridfinity base 7. grid dividers

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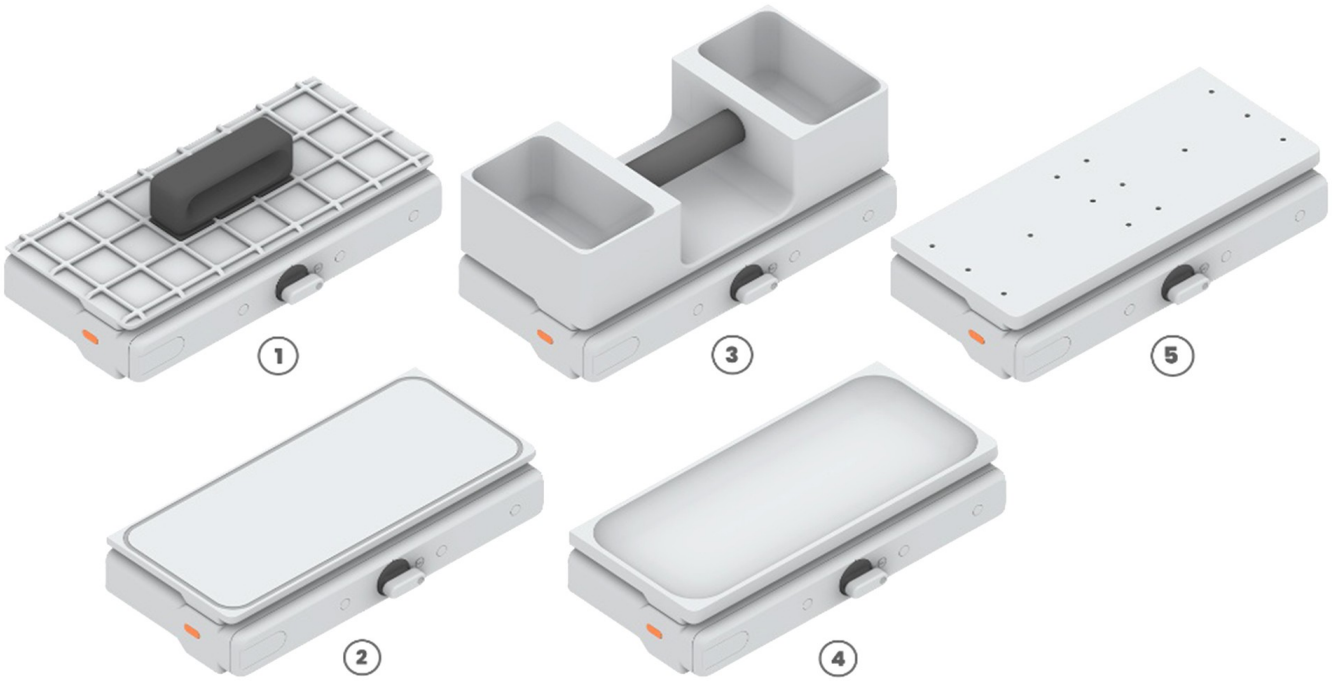
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Roof Types

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1. gridfinity 2. table 3. heavyweight 4. bowl 5. under table mount

·· 10

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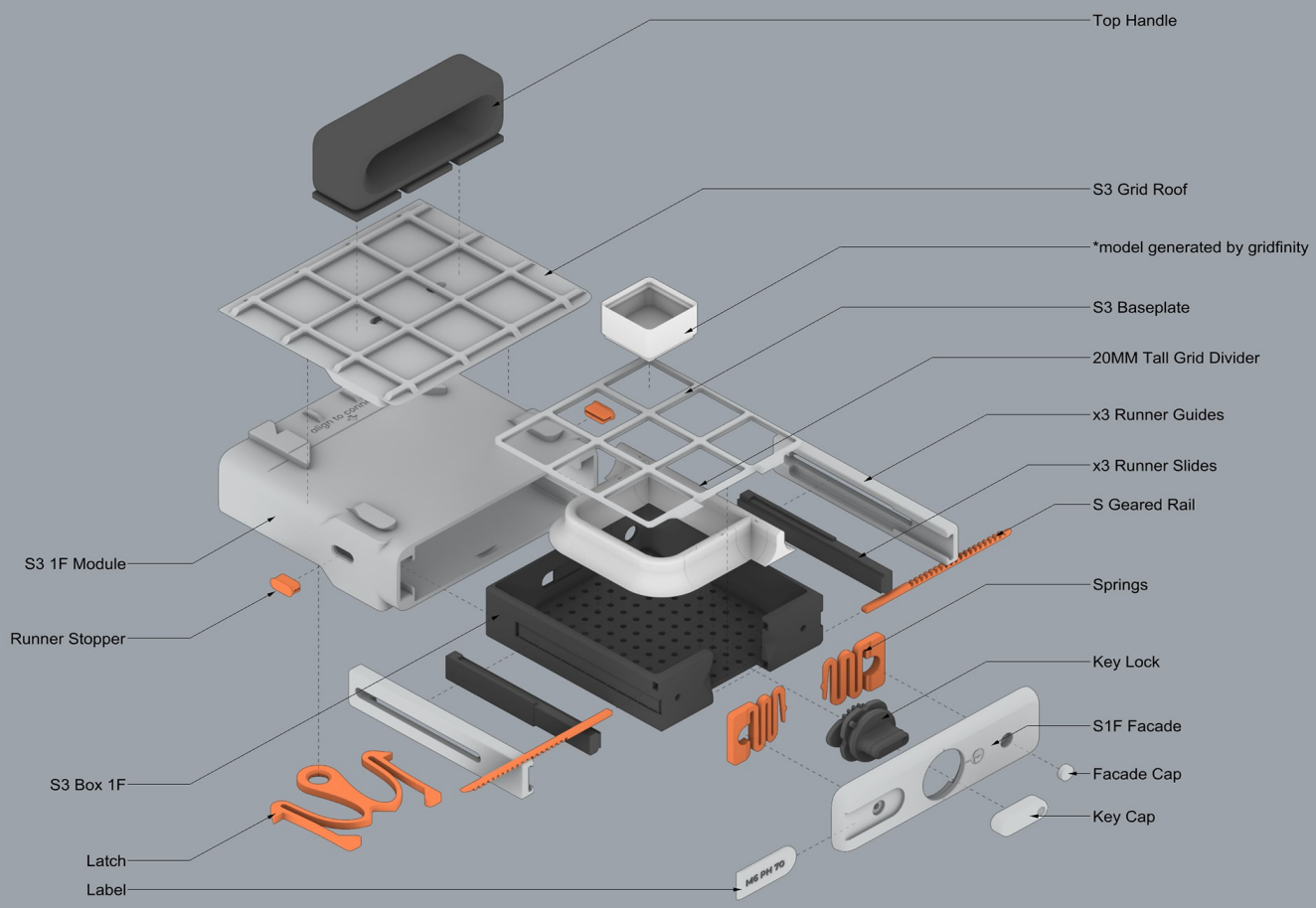
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Drawers Naming

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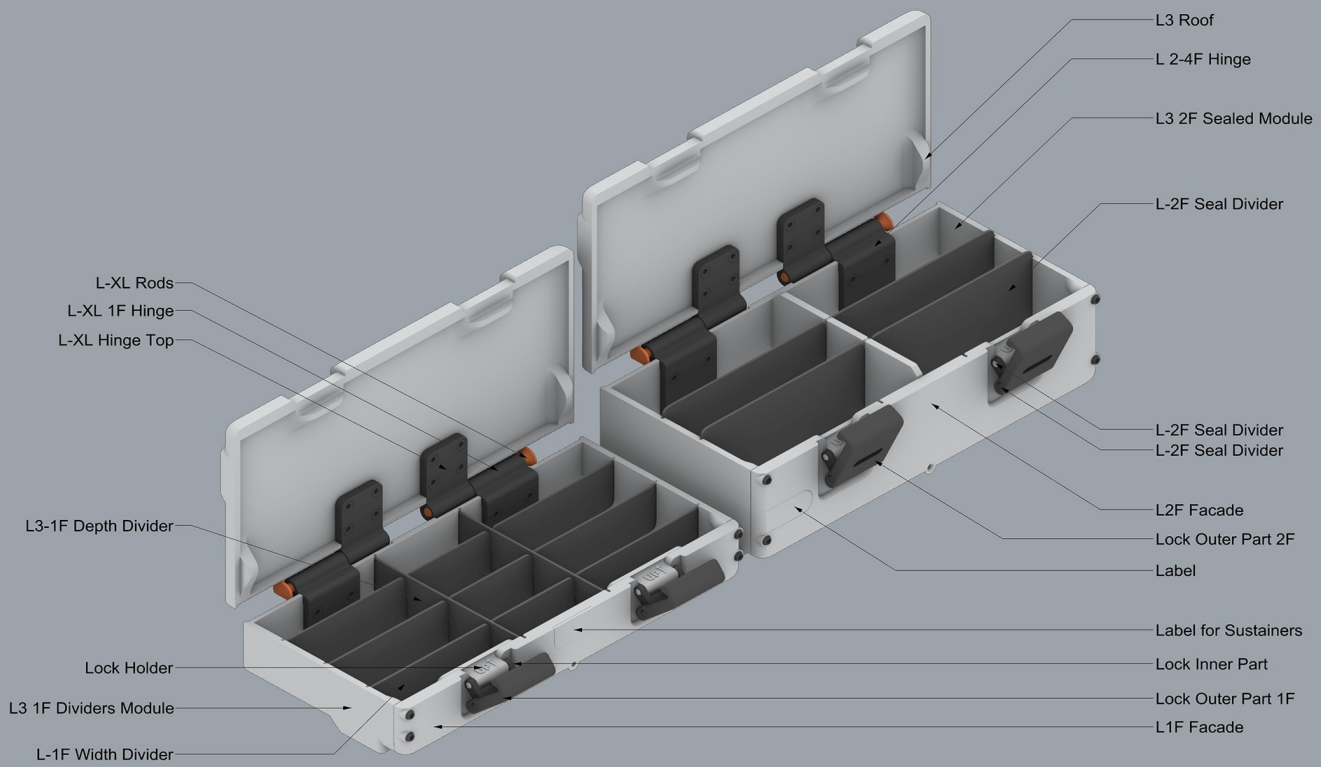
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Sustainers Naming

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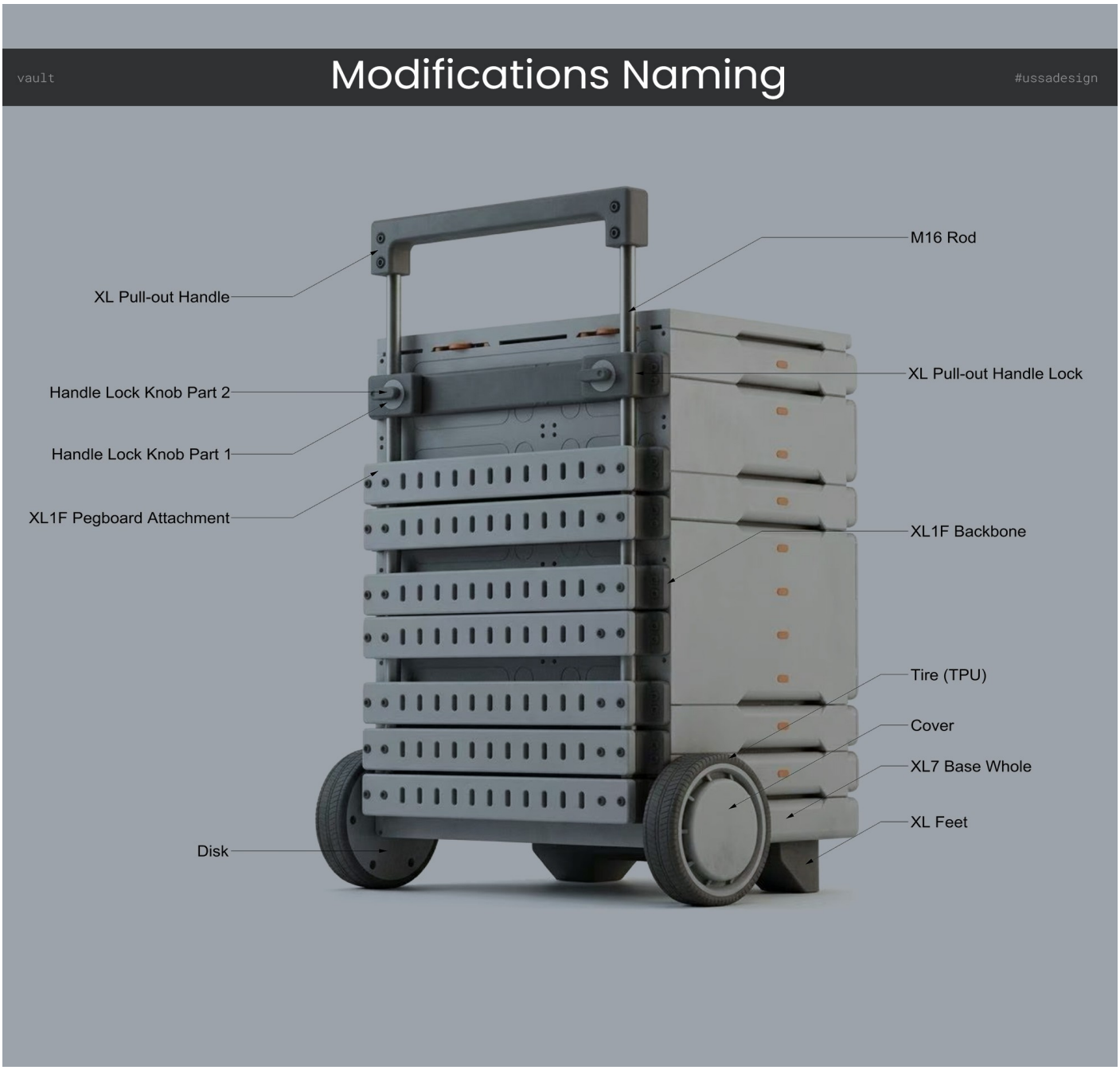


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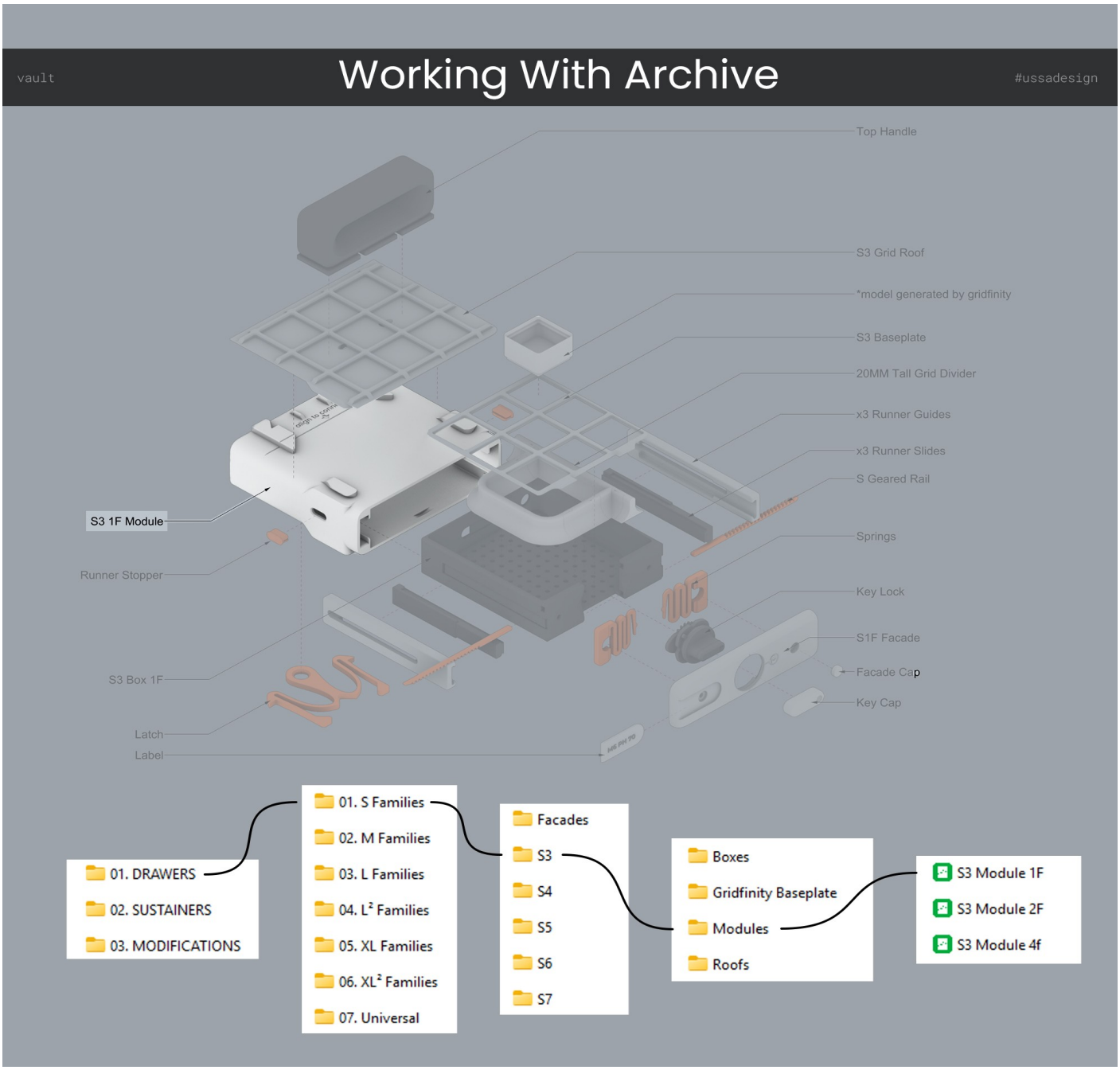
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Modifications Naming
#ussadesign
XL Pull-out Handle
Handle Lock Knob Part 2
Handle Lock Knob Part 1
XL1F Pegboard Attachment
Disk
M16 Rod
XL Pull-out Handle Lock
XL1F Backbone
Tire (TPU)
Cover
XL7 Base Whole
XL Feet

·· 13 ·····

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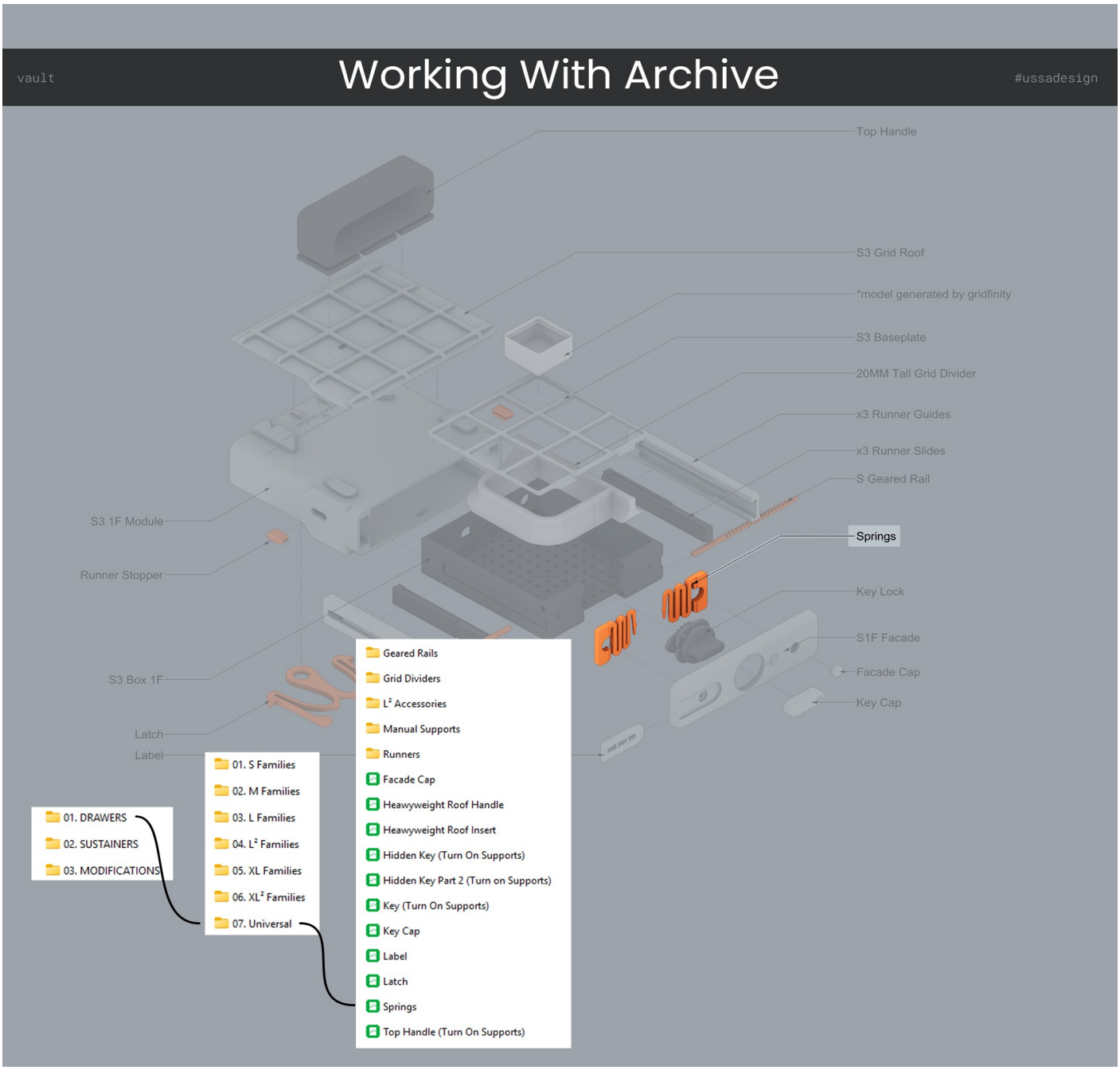
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vault
Working With Archive
#ussadesign
Top Handle
S3 Grid Roof
*model generated by gridfinity
S3 Baseplate
20MM Tall Grid Divider
x3 Runner Guides
x3 Runner Slides
S Geared Rail
Springs
Key Lock
S1F Facade
Facade Cap
Key Cap
S3 1F Module
Runner Stopper
S3 Box 1F
Latch
Label
01. DRAWERS
02. SUSTAINERS
03. MODIFICATIONS
01. S Families
02. M Families
03. L Families
04. L² Families
05. XL Families
06. XL² Families
07. Universal
Facades
S3
S4
S5
S6
S7
Boxes
Gridfinity Baseplate
Modules
Roofs
S3 Module 1F
S3 Module 2F
S3 Module 4f

·· 14 ·····

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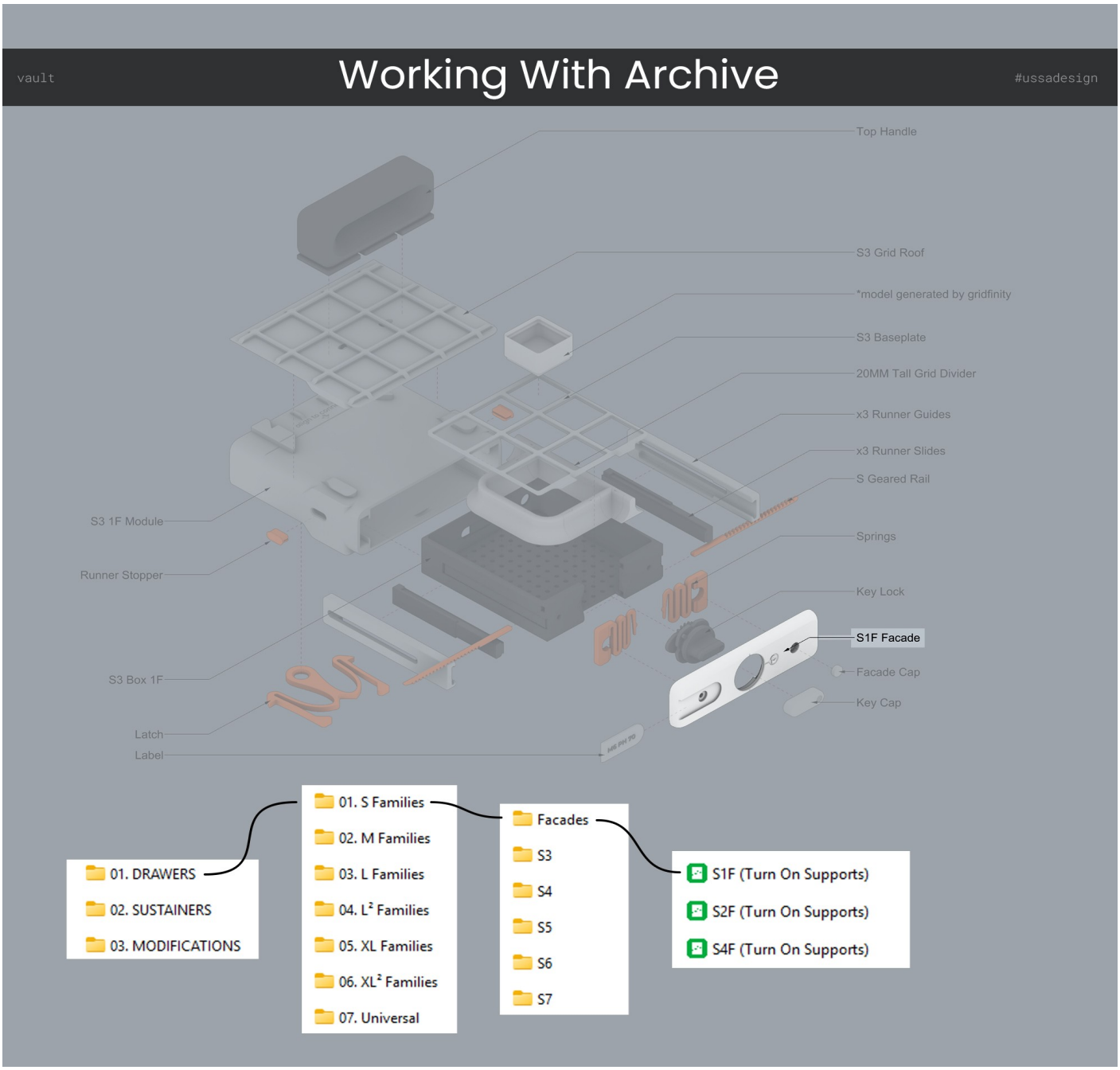
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vault
Working With Archive
#ussadesign
Top Handle
S3 Grid Roof
*model generated by gridfinity
S3 Baseplate
20MM Tall Grid Divider
x3 Runner Guides
x3 Runner Slides
S Geared Rail
Springs
Key Lock
S1F Facade
Facade Cap
Key Cap
S3 1F Module
Runner Stopper
S3 Box 1F
Latch
Label
01. DRAWERS
02. SUSTAINERS
03. MODIFICATIONS
01. S Families
02. M Families
03. L Families
04. L² Families
05. XL Families
06. XL² Families
07. Universal
Geared Rails
Grid Dividers
L² Accessories
Manual Supports
Runners
Facade Cap
Heawyweight Roof Handle
Heawyweight Roof Insert
Hidden Key (Turn On Supports)
Hidden Key Part 2 (Turn on Supports)
Key (Turn On Supports)
Key Cap
Label
Latch
Springs
Top Handle (Turn On Supports)

·· 15 ·····

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vault
Working With Archive
#ussadesign
Top Handle
S3 Grid Roof
*model generated by gridfinity
S3 Baseplate
20MM Tall Grid Divider
x3 Runner Guides
x3 Runner Slides
S Geared Rail
Springs
Key Lock
S1F Facade
Facade Cap
Key Cap
S3 1F Module
Runner Stopper
S3 Box 1F
Latch
Label
01. DRAWERS
02. SUSTAINERS
03. MODIFICATIONS
01. S Families
02. M Families
03. L Families
04. L² Families
05. XL Families
06. XL² Families
07. Universal
Facades
S3
S4
S5
S6
S7
S1F (Turn On Supports)
S2F (Turn On Supports)
S4F (Turn On Supports)

·· 16 ·····

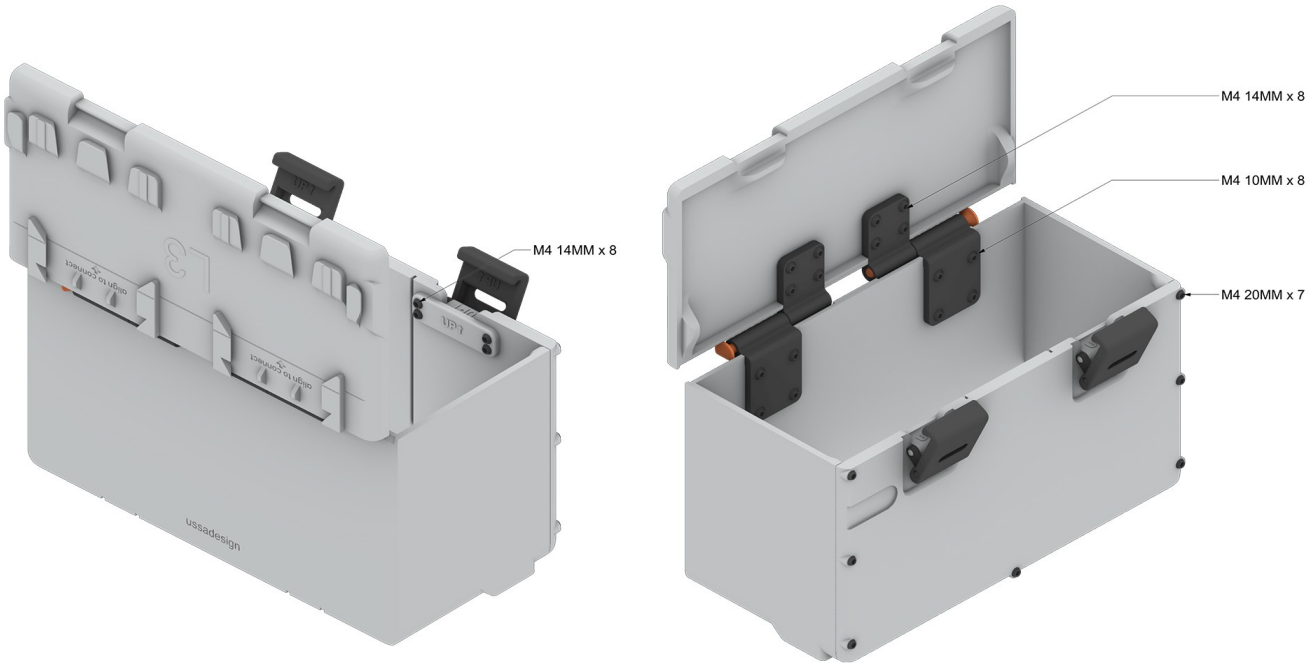
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vault

Required Hardware - Sustainers

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*x16 M4 14mm bolts, x8 M4 10mm bolts, x7 M4 20mm bolts - per module

·· 17

····· (Per-Case)

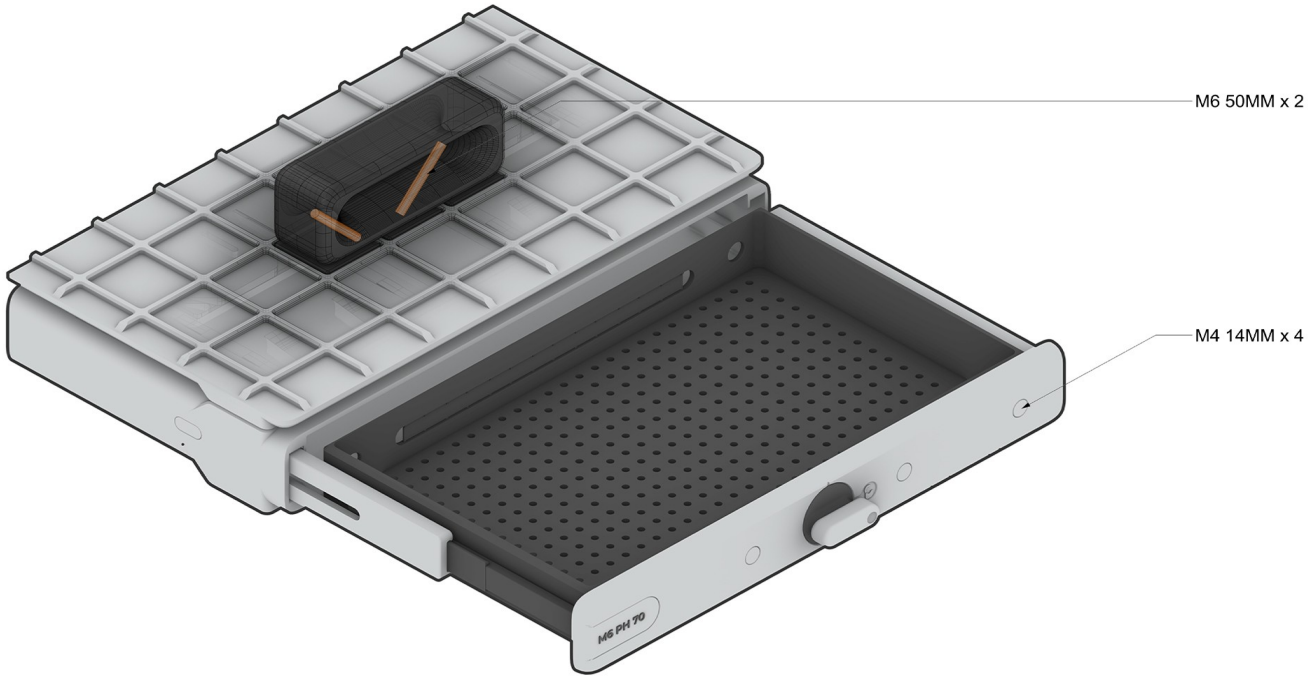
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Required Hardware - Drawers

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*x4 M4 14mm bolts - for 1F, x6 for 2F/4F

·· 18 ····· (Per-Drawer)

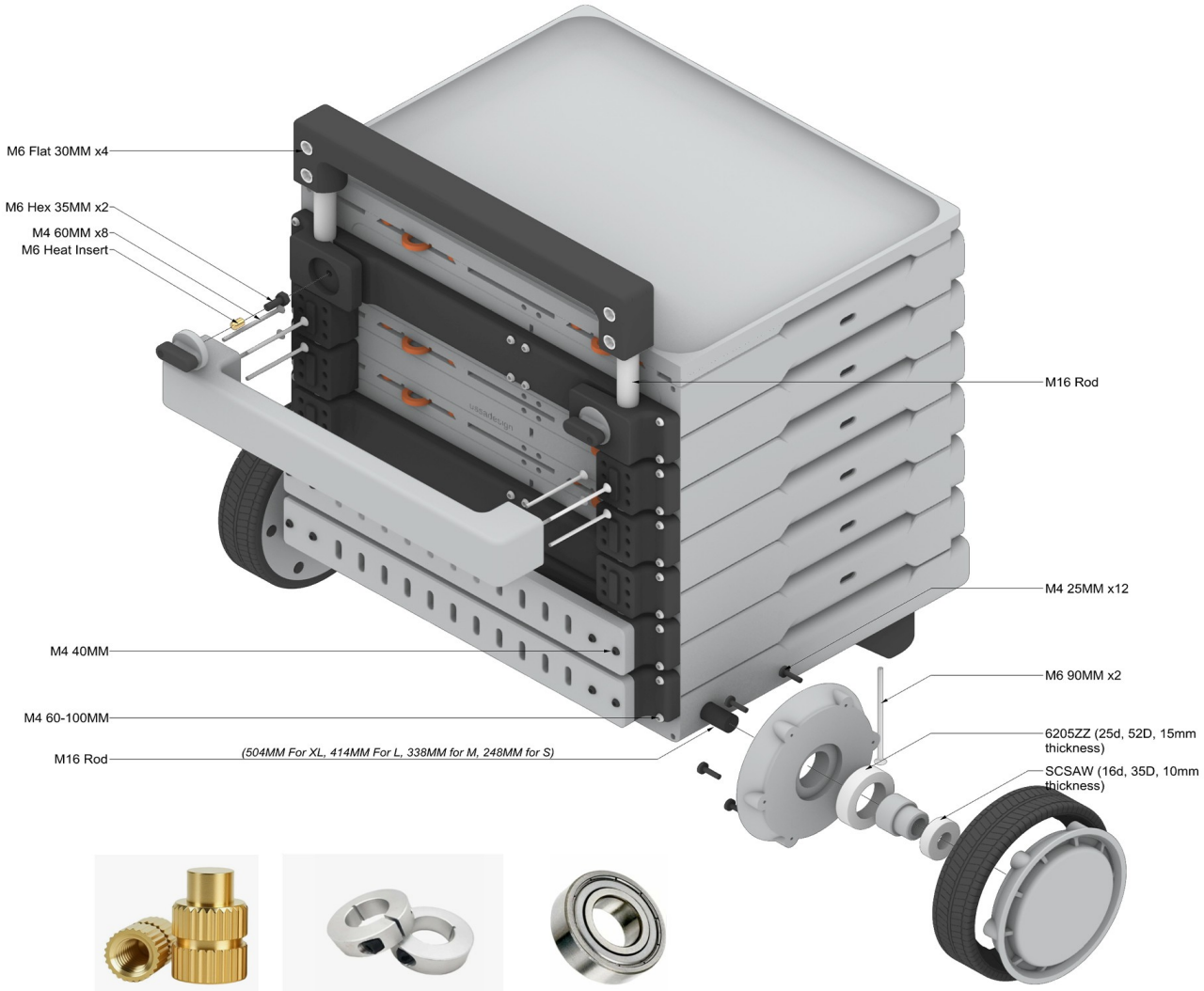
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Required Hardware - Backbone

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heat insert

SCSAW (16d, 35D, 10mm thickness)

6205ZZ (25d, 52D, 15mm thickness)

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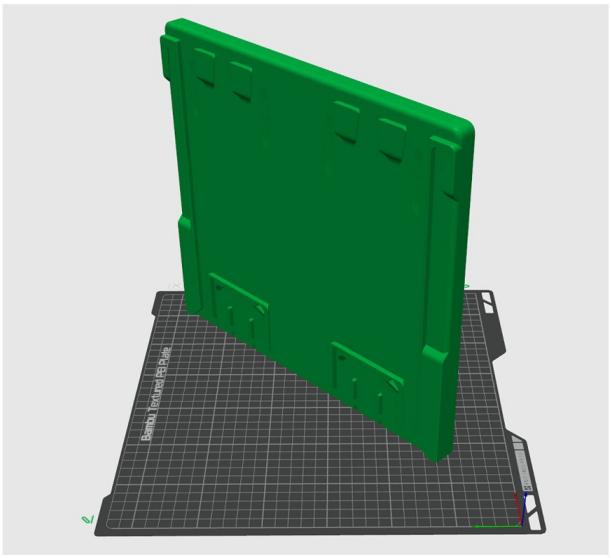
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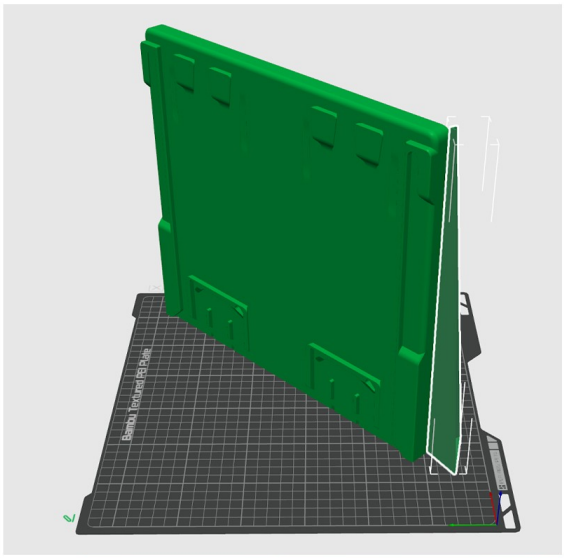
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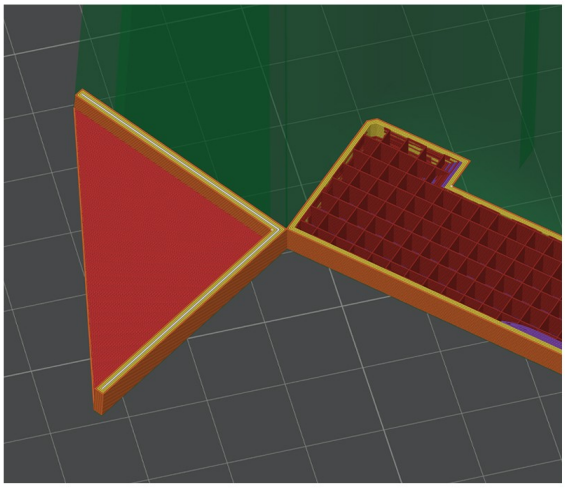
Manual Supports

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you can add manual supports for a more stable print experience, especially for tall prints

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